`pelib.mos_simple_2th`
A simplified power MOSFET loss model in Simscape Language

Documentation

Revision 1.0

Abstract

This document describes a Simscape Language component for converter-level loss and junction-temperature estimation of power MOSFETs. The switching transient is not resolved: the device commutates in one solver step and the switching energies are taken from single datasheet values, scaled to the operating point. Conduction and switching losses are separated between the transistor die and the diode die and delivered to two thermal ports. The model follows the same structure as the corresponding IGBT model, from which it differs in the treatment of the bidirectional channel, in the attribution of the third-quadrant losses, and in the option of a single junction when the diode is intrinsic.

Contents

1 Modelling assumptions

The model answers one question: how much power is dissipated, and where. It does not answer any question about the shape of the transient. The assumptions are listed explicitly, because most of the criticism that can be made of the model is already contained in them.

1. **The switching transient is not resolved.** There is no $\mathrm{d}i/\mathrm{d}t$, no voltage overshoot, no gate-loop dynamics. The channel conductance changes discontinuously with the gate command. This is deliberate: an instantaneous transition contributes exactly zero crossover energy to the conduction term $v\,i_{\mathrm{ch}}$, which leaves the switching energies entirely to the event mechanism of Section 4 and avoids double counting.

2. **Switching energies come from a single reference point.** One value of E_{on}, E_{off} and E_{rr} at $(I_{\mathrm{ref}}, V_{\mathrm{ref}})$, scaled by a power law. The error grows away from the reference point.

3. **No temperature dependence of the switching energies**, and none of the diode forward characteristic. For SiC this is a minor omission, since E_{on} and E_{off} are nearly temperature independent; for silicon it is not, and the parameters must be taken at the maximum expected junction temperature.

4. **Electrothermal feedback is optional and limited to $R_{\mathrm{ds,on}}$.** With $\alpha_R = 0$ the model is a pure feed-forward loss calculator and the electrical solution is independent of the thermal network. This is the default, and it is the configuration intended for nominal-value dimensioning.

5. **The gate command is a clean logic signal with dead time.** Because the channel switches discontinuously, cross-conduction is prevented by the external modulator, not by the device model.

2 Interface

2.1 Ports

Port	Type	Meaning
`p`	electrical	Drain
`n`	electrical	Source
`G`	physical signal in	Gate command, 0 or 1
`Ht`	thermal	MOSFET die
`Hd`	thermal	Diode die
`Pt`	physical signal out	Transistor dissipated power, W
`Pd`	physical signal out	Diode dissipated power, W

Table 1: Component interface. The port-side annotation in the `.ssc` source accepts only `left` and `right`.

2.2 Parameters

Name	Default	Unit	Meaning
`Rds_on`	2.5×10^{-3}	Ω	On-state resistance at `Tj_ref`
`alpha_R`	0	1/K	Temperature coefficient of $R_{\mathrm{ds,on}}$; 0 disables the feedback
`Tj_ref`	398.15	K	Reference junction temperature (125 °C)
`Roff`	1×10^{7}	Ω	Off-state resistance
`Vf`	1.0	V	Diode knee voltage
`Rd`	2.5×10^{-3}	Ω	Diode differential resistance
`two_die`	1	–	1: diode losses to `Hd`; 0: to `Ht`
`Eon_ref`	12×10^{-3}	J	Turn-on energy at the reference point
`Eoff_ref`	8×10^{-3}	J	Turn-off energy at the reference point
`Err_ref`	0.5×10^{-3}	J	Diode recovery energy at the reference point
`Iref`	600	A	Reference current of the energy data
`Vref`	600	V	Reference dc voltage of the energy data
`ki`	1	–	Current exponent
`kv`	1	–	Voltage exponent
`Vth_off`	50	V	Turn-off and recovery detection threshold
`Vth_on`	25	V	Turn-on detection threshold
`V_sm`	10	V	Smoothing width of the blocking indicator
`tau_s`	100×10^{-9}	s	Sample-and-hold time constant
`tau_v`	10×10^{-6}	s	Blocking-voltage estimator time constant
`tau_p`	50×10^{-6}	s	Energy-to-power lag time constant
`i_eps`, `v_eps`	1×10^{-3}	A, V	Smoothing constants

Name	Default	Unit	Meaning

Table 2: Parameters. The defaults are placeholders representative of a 1200 V/600 A SiC MOSFET half-bridge module at 125 °C; they are not taken from a specific datasheet.

3 Conduction

3.1 Two parallel branches

Let $v = v_\mathrm{p} - v_\mathrm{n}$ be the drain-source voltage and i the current entering the drain. Channel and diode are written as two branches across the same voltage, not as two alternative operating modes:

$$i_\mathrm{ch} = G_\mathrm{ch}\, v, \qquad G_\mathrm{ch} = \begin{cases} 1/R_\mathrm{ds} & G > 0.5 \\ 1/R_\mathrm{off} & \text{otherwise} \end{cases} \tag{1}$$

$$i_\mathrm{d} = \frac{1}{R_\mathrm{d}} \left[-v - V_\mathrm{f} \right]^+, \tag{2}$$

$$i = i_\mathrm{ch} - i_\mathrm{d}. \tag{3}$$

The positive part is regularised as

$$[x]^+ \;\simeq\; \frac{x + \sqrt{x^2 + \varepsilon^2}}{2}, \tag{4}$$

which removes the kink at $x = 0$ at the cost of a residual current $\varepsilon/(2R_\mathrm{d})$ at zero overdrive; with the default values this is 0.2 A, three orders of magnitude below the rated current. For $v \ll -V_\mathrm{f}$ the two terms of (4) cancel to $\varepsilon^2/(4|x|)$, so the blocking leakage is negligible and, more importantly, bounded.

This formulation is the main structural difference with respect to the IGBT model. In the third quadrant with the gate high, both branches are active simultaneously and the current splits according to their characteristics:

$$\frac{i_\mathrm{d}}{|i|} \;\neq\; 0 \quad \text{only if} \quad R_\mathrm{ds}\,|i| > V_\mathrm{f}. \tag{5}$$

With $R_\mathrm{ds} = 2.5\,\mathrm{m\Omega}$ and $V_\mathrm{f} = 1\,\mathrm{V}$ the diode starts to take a share above 400 A, which is the physical behaviour of a synchronously driven device and does not need to be imposed.

3.2 Loss attribution

$$P_\mathrm{c,t} = v\, i_\mathrm{ch} = \frac{v^2}{R_\mathrm{ds}} = R_\mathrm{ds} i_\mathrm{ch}^2, \qquad P_\mathrm{c,d} = -v\, i_\mathrm{d}. \tag{6}$$

Both are non-negative by construction, in either quadrant. The channel loss is charged to `Ht` regardless of the sign of the current: synchronous rectification is a transistor loss, not a diode loss. Note also that the transistor conduction loss is quadratic in current, whereas the IGBT model gives the affine form $V_\mathrm{ce,sat} i + R_\mathrm{on} i^2$. At partial load the MOSFET is therefore comparatively better, and a dimensioning based on the rated point alone will be conservative rather than optimistic.

3.3 Temperature dependence of $R_{\mathrm{ds,on}}$

Setting $\alpha_R \neq 0$ enables

$$R_{\mathrm{ds}} = R_{\mathrm{ds,on}}\Big[\,0.2 + \big[\,x_R - 0.2\,\big]^{+}\,\Big], \qquad x_R = 1 + \alpha_R\,(T_{\mathrm{Ht}} - T_{\mathrm{j}}^{\mathrm{ref}}), \tag{7}$$

with the positive part again regularised as in (4). The floor at $0.2\,R_{\mathrm{ds,on}}$ keeps the resistance positive if the thermal network is initialised far below the operating temperature; without it, a coefficient of 0.006/K would produce a negative resistance below −106 °C, which is outside any physical range but well inside the range a solver can visit during initialisation.

Enabling this term closes an algebraic loop between the electrical solution and the thermal network. The loop is well posed and Simscape handles it, but it enlarges the nonlinear system and it makes the losses depend on the initial temperature of the Foster network. For a dimensioning at nominal values the recommended configuration is $\alpha_R = 0$ with $R_{\mathrm{ds,on}}$ read at the maximum junction temperature. Representative coefficients around 125 °C are 0.005/K to 0.008/K for SiC and 0.008/K to 0.012/K for silicon superjunction devices.

4 Switching losses

4.1 Scaling law

Each event adds an energy

$$E = E_{\mathrm{ref}} \left(\frac{I}{I_{\mathrm{ref}}}\right)^{k_i} \left(\frac{V_{\mathrm{bl}}}{V_{\mathrm{ref}}}\right)^{k_v}, \tag{8}$$

where $k_i = k_v = 1$ reproduces the usual bilinear scaling. The power-law form is preferred to a polynomial fit because it remains positive for any argument, which matters when the solver evaluates the expression during a rejected step.

V_{bl} is not the instantaneous terminal voltage, which at the detection instant is only V_{th}. It is a held estimate of the blocking voltage, obtained by tracking v only while the device is blocking:

$$\tau_v\,\dot{V}_{\mathrm{bl}} = h_{\mathrm{b}}\,(v - V_{\mathrm{bl}}), \qquad h_{\mathrm{b}} = \tfrac{1}{2}\left[1 + \tanh\frac{v - V_{\mathrm{th,off}}}{V_{\mathrm{sm}}}\right]. \tag{9}$$

The tanh indicator is used instead of a conditional expression so that no additional zero crossing is generated. With $\tau_v \ll$ the blocking interval, V_{bl} converges to the dc-link voltage; when the voltage collapses, $h_{\mathrm{b}} \to 0$ and the value is frozen and available at the event.

4.2 Event detection

The switching instants are detected on the terminal voltage rather than on the gate command, because the terminal voltage is what distinguishes a commutation that actually happened from one that was only commanded.

Event	Trigger	Energy	Current
Turn-on	v falls below $V_{\mathrm{th,on}}$	$G \cdot E_{\mathrm{on}}$	terminal current at the event
Turn-off	v rises above $V_{\mathrm{th,off}}$	E_{off}	i_{th}, held
Recovery	v rises above $V_{\mathrm{th,off}}$	E_{rr}	i_{dh}, held

Table 3: Event table. Two distinct thresholds provide hysteresis and prevent chattering when the terminal voltage lingers near the detection level.

The factor G on the turn-on energy suppresses the spurious detection produced during the dead time, when the voltage collapses because the diode of this same device starts to conduct: there the gate is low and the term vanishes. A turn-on that occurs while the diode is already conducting produces no down-crossing at all, since v is already negative; a zero-voltage turn-on therefore costs nothing, without any special-case logic.

At the turn-on event the current is sampled directly, because the commutation from the complementary device is complete by the time the voltage has fallen below $V_{\mathrm{th,on}}$. At the turn-off event this is not true: the current has already left the device, and a held value is required.

4.3 Sample-and-hold states

Turn-off and reverse recovery share a single trigger, because the terminal voltage rises in two distinct situations: when this transistor turns off, and when the complementary switch turns on and reverse-biases this diode. Both energies are evaluated at every up-crossing, and the correct one is selected by two tracking states rather than by a mode test:

$$\tau_s \, \dot{i}_{\mathrm{th}} = G \left([\, i_{\mathrm{ch}} \,]^{+} - i_{\mathrm{th}} \right), \tag{10}$$

$$\tau_s \, \dot{i}_{\mathrm{dh}} = i_{\mathrm{d}} - i_{\mathrm{dh}}. \tag{11}$$

Equation (10) tracks the *positive part* of the channel current while the gate is high and holds when it is low. Two consequences follow. In first-quadrant conduction, i_{th} equals the load current and is frozen at the value it had immediately before the gate went low, which is exactly what E_{off} requires. In synchronous rectification the channel current is negative, $[\, i_{\mathrm{ch}} \,]^{+} \to 0$, and the state decays to zero: the subsequent up-crossing produces $E_{\mathrm{off}} = 0$, which is correct, since turning the channel off into its own conducting diode is a soft transition.

Equation (11) tracks the diode current unconditionally. While the channel shunts the diode, $i_{\mathrm{d}} \simeq 0$ and the state follows it down; during the dead time the diode takes the full load current and the state recovers it. Hence exactly one E_{rr} event per switching period, in the half period in which the diode actually conducts, and $E_{\mathrm{rr}} = 0$ in the other half.

The choice of τ_s is the one real constraint of the scheme:

$$\tau_s \ll t_{\mathrm{dead}}, \tag{12}$$

otherwise i_{dh} does not reach the load current before the recovery event and E_{rr} is underestimated. The default 100 ns against a dead time of 1 µs to 2 µs gives ten to twenty time constants, i.e. an error below 0.01 %. Reducing τ_s further only adds a fast mode to the system without improving the result.

4.4 Sequence over one switching period

Figure 1 shows the two half periods of a leg operating with positive and with negative load current, and the resulting event attribution.

5 From energy to power

The event energies are accumulated in two event variables E_{q}, which are piecewise constant and jump at each event. A continuous state follows them through a first-order lag:

$$\tau_p \, \dot{E}_{\mathrm{o}} = E_{\mathrm{q}} - E_{\mathrm{o}}, \qquad P_{\mathrm{sw}} = \frac{E_{\mathrm{q}} - E_{\mathrm{o}}}{\tau_p} = \dot{E}_{\mathrm{o}}. \tag{13}$$

Since the delivered power is by construction the derivative of E_{o},

$$\int_0^t P_{\mathrm{sw}} \, \mathrm{d}\tau = E_{\mathrm{o}}(t) = E_{\mathrm{q}}(t) - \tau_p P_{\mathrm{sw}}(t), \tag{14}$$

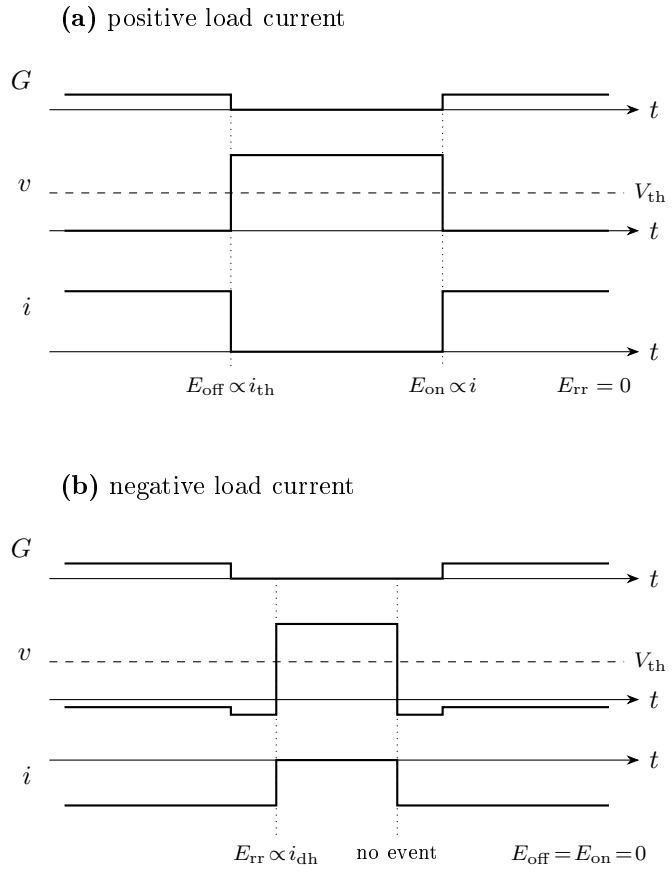


Figure 1: Attribution of the switching energies. In (a) the channel conducts the positive load current: the up-crossing of v charges E_{off} with the held channel current, and the down-crossing charges E_{on} with the freshly commutated terminal current; the diode never conducts, so $i_{\mathrm{dh}} = 0$ and no recovery energy is produced. In (b) the current is negative: the channel conducts synchronously, i_{th} has decayed to zero, the diode picks up the current during the dead time and the up-crossing produced by the turn-on of the complementary switch charges E_{rr} only. Waveforms are schematic; the dead-time intervals are exaggerated.

so the energy is conserved exactly up to the residual $\tau_p P_{\mathrm{sw}}$, which is bounded by the largest single event energy and vanishes in the mean. The lag introduces no bias: it only spreads each pulse over a time τ_p.

The choice of τ_p follows from two inequalities:

$$T_{\mathrm{sw}} \lesssim \tau_p \ll \min_k \tau_{\mathrm{th},k}, \tag{15}$$

where $\tau_{\mathrm{th},k}$ are the time constants of the Foster network. The lower bound smooths the pulse train and keeps the solver from tracking a sequence of impulses; the upper bound guarantees that the thermal response is not distorted. At 20 kHz and with a fastest Foster time constant of the order of 1 ms, any τ_p between 50 µs and 200 µs is acceptable.

6 Thermal ports

6.1 Sign convention

The thermal branches are declared as

```
Qt :  Ht.Q -> *;     Qd :  Hd.Q -> *;
```

so that Q is positive when heat flows into the component. The equations therefore read

$$Q_{\mathrm{t}} = -P_{\mathrm{t}}, \qquad Q_{\mathrm{d}} = -P_{\mathrm{d}}, \tag{16}$$

with P_{t} and P_{d} the dissipated powers, also exported as physical signal outputs for bookkeeping.

6.2 One die or two

$$P_{\mathrm{t}} = P_{\mathrm{c,t}} + P_{\mathrm{sw,t}} + (1-\nu)\,(P_{\mathrm{c,d}} + P_{\mathrm{sw,d}}), \qquad P_{\mathrm{d}} = \nu\,(P_{\mathrm{c,d}} + P_{\mathrm{sw,d}}), \tag{17}$$

where $\nu = \texttt{two_die} \in \{0, 1\}$.

With $\nu = 1$ the diode losses are delivered to `Hd`. This is the case of a module carrying a separate antiparallel die, typically a SiC MOSFET with a SiC Schottky diode, where the two junctions have different $R_{\mathrm{th(j-c)}}$ and deserve separate temperatures.

With $\nu = 0$ the diode losses are added to `Ht` instead. This must be used whenever the diode is the intrinsic body diode: it is the same die as the channel, and modelling two independent junctions would be an error rather than a refinement. Port `Hd` then carries no heat, but it must still be connected, for instance to a Thermal Reference or to the same node as `Ht`.

A practical consequence: since the body diode of a synchronously driven MOSFET conducts only during the dead time, its contribution is small, but it is concentrated on the same junction that already carries the channel loss. The single-junction configuration is therefore not conservative by accident; it is simply correct.

7 Parameter extraction

- $R_{\mathrm{ds,on}}$ at the junction temperature at which the dimensioning is performed and at the actual gate voltage. Datasheets normally give 25 °C and 150 °C; interpolate, and remember that the value at 25 °C is the marketing number, not the design number.
- V_{f} and R_{d} from the third-quadrant characteristic measured with V_{gs} at its negative off-state value, not from the synchronous-conduction curve, which is the channel and is already accounted for by (1).
- E_{on}, E_{off} from the energy-versus-current curves at the gate voltage, gate resistance and dc voltage of the application. Choose I_{ref} close to the current at which the converter actually operates: with $k_i = 1$ the error of (8) is zero at the reference point and grows away from it.
- E_{rr} from the recovery-energy curve, or from $E_{\mathrm{rr}} \simeq V_{\mathrm{dc}} Q_{\mathrm{rr}}$ when only Q_{rr} is given. For a Schottky diode the recovered charge is capacitive only: the appropriate value is small but not zero.

Two remarks specific to MOSFETs:

1. The datasheet E_{on} is measured in a hard-switched double-pulse test and already includes the reverse recovery of the complementary diode and the discharge of C_{oss}. Neither must be added again.
2. Under zero-voltage switching the datasheet E_{on} is meaningless. If the model is used in a resonant, dual-active-bridge or phase-shifted converter, E_{on} must be replaced by the C_{oss} hysteresis energy of the device, one to two orders of magnitude smaller. It is worth checking, by monitoring `Pt`, that the model is not detecting turn-on events that in the real converter are soft.

8 Limits of validity

- No transient waveform information: no $\mathrm{d}i/\mathrm{d}t$, no overshoot, no gate resistance dependence, no snubber evaluation. For those, a switching-dynamics model such as `mos_dyn` is required.
- Energies scaled from one reference point; error grows with distance from it, and the linear current law underestimates the curvature of the real $E(I)$ characteristic at low current.
- No temperature dependence of the switching energies or of the diode characteristic.
- Off-state leakage is set by R_{off}, not by a physical model. At 1200 V the default gives 0.14 W of spurious blocking loss, negligible against the hundreds of watts of a rated device but worth remembering if the model is used at very light load.
- The model relies on external dead time. With zero dead time the recovery current is underestimated, and cross-conduction is not prevented by the component.

9 Suggested verification

1. Constant duty cycle and constant load current: compare $\overline{P_{\mathrm{t}}} + \overline{P_{\mathrm{d}}}$ against the hand calculation $R_{\mathrm{ds}} I_{\mathrm{rms}}^2 + (E_{\mathrm{on}} + E_{\mathrm{off}}) f_{\mathrm{sw}} (I/I_{\mathrm{ref}})(V_{\mathrm{dc}}/V_{\mathrm{ref}})$.
2. Sweep the sign of the load current and verify that P_{d} collapses in the half period in which the channel conducts synchronously, and that E_{off} vanishes there.
3. Set $\nu = 0$ and verify that P_{t} gains exactly what P_{d} loses.
4. If the solver reports scaling problems, add explicit `nominal` values to the energy variables `Eqt`, `Eqd`, `Eot`, `Eod`.

A Source listing

```
component mos_simple_2th
% MOS_SIMPLE_2TH Simplified power MOSFET + antiparallel diode, two thermal ports
%
% Loss-oriented switch model. The switching transient is not resolved: the
% device commutates in one solver step and the switching energies come from
% single datasheet values scaled to the actual current and dc-link voltage.
% Losses are delivered to two thermal ports:
%
% Ht : MOSFET die -> channel conduction (v*i_ch, both quadrants) + Eon + Eoff
% Hd : diode die -> diode conduction (-v*i_d) + Err
%
% Differences with respect to IGBT_SIMPLE_2TH
% -------------------------------------------
% 1. Conduction is purely resistive (Rds_on), no knee voltage, and the channel
% is bidirectional. With the gate on and negative current the current
% returns through the channel (synchronous rectification) and the loss is
% charged to Ht, not to Hd. Channel and diode are two parallel branches on
% the same terminal voltage, so the current split in the third quadrant
% comes out of the equations, not out of a mode selection.
% 2. Rds_on can be made temperature dependent (alpha_R ~= 0), which closes the
% electrothermal loop through Ht. With alpha_R = 0 (default) the loop is
% absent and the model behaves like the IGBT one: fixed 125 degC parameters.
% 3. The diode can be a separate die (SiC MOSFET + antiparallel SiC Schottky)
% or the intrinsic body diode of the same die. Parameter two_die selects it:
% two_die = 1 -> diode losses to Hd (separate die)
% two_die = 0 -> diode losses to Ht (body diode, one junction)
% With two_die = 0 the port Hd carries no heat but must still be connected,
% e.g. to a Thermal Reference or to the same node as Ht.
%
% Event logic (same scheme as the IGBT model)
```

```
% ------------------------------------------
% Turn-on : detected as the terminal voltage falling below Vth_on. The energy
% is charged only if the gate is high, so the voltage collapse
% caused by the diode starting to conduct during the dead time
% produces no Eon. The current is sampled at the event, when the
% commutation from the complementary device is already complete.
% Turn-off : detected as the terminal voltage rising above Vth_off. The same
% crossing occurs when the diode of THIS device is reverse biased by
% the turn-on of the complementary switch, so both Eoff and Err are
% evaluated at that event and the correct one is selected by the two
% sample-and-hold states:
% ith : tracks the positive part of the channel current while the
% gate is high, holds while the gate is low. It is zero
% during synchronous rectification, so Eoff = 0 there.
% idh : tracks the diode current. It is ~0 while the channel
% shunts the diode, and recovers the load current during the
% dead time, so Err is charged once per switching period in
% the half period where the diode actually conducts.
% Energies are converted to power by a first-order lag of time constant tau_p,
% which conserves energy exactly: int(P dt) = sum of the event energies.
%
% Default parameters are placeholders representative of a 1200 V / 600 A SiC
% MOSFET half-bridge module at Tj = 125 degC, with an antiparallel SiC Schottky
% diode. Replace them with datasheet values before use.
% For a module without a separate diode (body diode only) set
% two_die = 0, Vf ~ 3.2 V, Rd ~ 3 mOhm and Err from Qrr of the body diode.

nodes
    p = foundation.electrical.electrical; % D:left
    n = foundation.electrical.electrical; % S:right
    Ht = foundation.thermal.thermal; % Ht:left
    Hd = foundation.thermal.thermal; % Hd:right
end

inputs
    G = {0, '1'}; % G:left
end

outputs
    Pt = {0, 'W'}; % Pt:right
    Pd = {0, 'W'}; % Pd:right
end

parameters
    % --- conduction, transistor ---
    Rds_on = {2.5e-3, 'Ohm'}; % On-state resistance at Tj_ref
    alpha_R = {0, '1/K'}; % Rds_on temperature coefficient (0 disables it)
    Tj_ref = {398.15, 'K' }; % Reference junction temperature (125 degC)
    Roff = {1e7, 'Ohm'}; % Off-state resistance
    % --- conduction, diode ---
    Vf = {1.0, 'V' }; % Diode knee voltage at Tj_ref
    Rd = {2.5e-3, 'Ohm'}; % Diode differential resistance at Tj_ref
    two_die = {1, '1' }; % 1 = separate diode die (Hd), 0 = body diode (Ht)
    % --- switching energies ---
    Eon_ref = {12e-3, 'J' }; % Turn-on energy at Iref, Vref
    Eoff_ref = {8e-3, 'J' }; % Turn-off energy at Iref, Vref
    Err_ref = {0.5e-3, 'J' }; % Diode recovery energy at Iref, Vref
    Iref = {600, 'A' }; % Reference current of the energy data
    Vref = {600, 'V' }; % Reference dc voltage of the energy data
    ki = {1, '1' }; % Current exponent
    kv = {1, '1' }; % Voltage exponent
    % --- detection and filtering ---
    Vth_off = {50, 'V' }; % Turn-off / recovery detection threshold
    Vth_on = {25, 'V' }; % Turn-on detection threshold (< Vth_off)
    V_sm = {10, 'V' }; % Smoothing width of the blocking detector
    tau_s = {100e-9, 's' }; % Sample-and-hold time constant (<< dead time)
    tau_v = {10e-6, 's' }; % Blocking voltage estimator time constant
    tau_p = {50e-6, 's' }; % Energy-to-power lag time constant
    i_eps = {1e-3, 'A' }; % Current smoothing
    v_eps = {1e-3, 'V' }; % Voltage smoothing
end

variables
```

```
    v = {0, 'V' }; % Terminal voltage, drain to source
    ich = {0, 'A' }; % Channel current
    ichp = {0, 'A' }; % Positive part of the channel current
    idio = {0, 'A' }; % Diode forward current
    vfd = {0, 'V' }; % Diode forward overdrive, -v - Vf
    ith = {0, 'A' }; % Held channel current, used for Eoff
    idh = {0, 'A' }; % Held diode current, used for Err
    vbl = {600, 'V' }; % Held blocking voltage, used for all energies
    hb = {0, '1' }; % Blocking indicator, 0..1
    Gch = {0, 'S' }; % Channel conductance
    Rds = {1e-3,'Ohm'}; % Actual on-state resistance
    xr = {1, '1' }; % Rds_on temperature factor
    Eot = {0, 'J' }; % Lagged transistor energy
    Eod = {0, 'J' }; % Lagged diode energy
    Pswt = {0, 'W' }; % Transistor switching power
    Pswd = {0, 'W' }; % Diode switching power
    Pct = {0, 'W' }; % Transistor conduction power
    Pcd = {0, 'W' }; % Diode conduction power
end

variables (Event = true)
    Eqt = {0, 'J'}; % Cumulative transistor switching energy
    Eqd = {0, 'J'}; % Cumulative diode switching energy
end

branches
    i : p.i -> n.i;
    Qt : Ht.Q -> *;
    Qd : Hd.Q -> *;
end

equations
    assert(Rds_on > 0)
    assert(Rd > 0)
    assert(Roff > Rds_on)
    assert(Vth_on < Vth_off)
    assert(tau_s > 0)
    assert(tau_p > 0)

    v == p.v - n.v;

    % ---------------- on-state resistance ----------------
    if alpha_R == {0, '1/K'}
        xr == 1;
        Rds == Rds_on;
    else
        xr == 1 + alpha_R*(Ht.T - Tj_ref);
        Rds == Rds_on*(0.2 + (xr - 0.2 + sqrt((xr - 0.2)^2 + 0.01))/2);
    end

    % ---------------- channel and diode ----------------
    if G > 0.5
        Gch == 1/Rds;
    else
        Gch == 1/Roff;
    end
    ich == Gch*v;

    vfd == -v - Vf;
    idio == (vfd + sqrt(vfd^2 + v_eps^2))/(2*Rd);

    i == ich - idio;

    % ---------------- sample and hold ----------------
    ichp == (ich + sqrt(ich^2 + i_eps^2))/2;
    tau_s*ith.der == G*(ichp - ith);
    tau_s*idh.der == idio - idh;

    hb == (1 + tanh((v - Vth_off)/V_sm))/2;
    tau_v*vbl.der == hb*(v - vbl);

    % ---------------- energy to power ----------------
    tau_p*Eot.der == Eqt - Eot;
```

```
    tau_p*Eod.der == Eqd - Eod;
    Pswt == (Eqt - Eot)/tau_p;
    Pswd == (Eqd - Eod)/tau_p;

    % ---------------- losses and thermal ports ----------------
    Pct == v*ich;
    Pcd == -v*idio;

    Pt == Pct + Pswt + (1 - two_die)*(Pcd + Pswd);
    Pd == two_die*(Pcd + Pswd);

    Qt == -Pt;
    Qd == -Pd;
end

events
    % Turn-on: terminal voltage collapse with the gate high.
    when edge(v < Vth_on)
        Eqt = Eqt + G*Eon_ref*(((i + abs(i))/(2*Iref))^ki)*((abs(vbl)/Vref)^kv);
    end

    % Turn-off of the channel, or reverse recovery of the diode of this device.
    when edge(v > Vth_off)
        Eqt = Eqt + Eoff_ref*((ith/Iref)^ki)*((abs(vbl)/Vref)^kv);
        Eqd = Eqd + Err_ref*((idh/Iref)^ki)*((abs(vbl)/Vref)^kv);
    end
end

end
```